

PUGA DIGITAL TECHNOLOGIES LTD

PugaAI Health

Your AI-powered voice health navigator

Product Requirements Document (PRD)

Version: 1.0

Status: Development Specification

Product: PugaAI Health

Parent Ecosystem: Puga TriniCare

Primary AI Technology for NAIC MVP: N-ATLAS

Document Audience: Product, Engineering, AI/ML, UI/UX, QA, Security and Technical Leadership Teams

1. Executive Summary

PugaAI Health is an AI-powered, voice-first health access and navigation platform within the Puga TriniCare ecosystem.

It enables users to interact with PugaAI using natural voice or text to:

- obtain trusted primary-health information;
- ask health-related questions;
- navigate appropriate healthcare services;
- identify when professional care may be appropriate;
- find participating healthcare facilities and providers;
- initiate appointments and other healthcare services;
- access Puga TriniCare services;
- interact through smartphone/web interfaces and, through PugaAccess, low-bandwidth channels such as IVR, USSD and SMS.

For the NAIC implementation, PugaAI Health will integrate N-ATLAS as a core component of the voice-first experience, particularly for speech recognition and Nigerian-language interaction.

PugaAI Health is not intended to function as an autonomous doctor or independently diagnose, prescribe or make irreversible clinical decisions.

Its initial role is:

Understand → Inform → Navigate → Connect → Assist

2. Product Vision

Vision

To make trusted healthcare information and healthcare access available through natural human conversation, regardless of a user's digital literacy, device type or connectivity level.

Long-Term Vision

PugaAI Health should become the conversational intelligence layer of Puga TriniCare, connecting patients and other authorized users to healthcare information, services, providers and transactions.

3. Product Positioning

Product Name

PugaAI Health

Tagline

Your AI-powered voice health navigator

Product Category

AI-powered voice health access, information and navigation platform.

Core Proposition

Speak naturally to PugaAI Health, understand your health-information needs, receive appropriate guidance and, when necessary, connect to the right healthcare service within the Puga TriniCare ecosystem.

4. Problem Statement

Healthcare access in many Nigerian communities remains constrained by fragmented healthcare services, limited digital literacy, language barriers, smartphone and connectivity requirements, and difficulty determining where and how to obtain appropriate care.

Existing digital-health platforms frequently require users to navigate menus, type queries, understand medical terminology or use smartphone applications.

PugaAI Health addresses this access gap by allowing users to communicate naturally through voice and text while connecting the AI interaction to actual healthcare services.

5. Target Users

Primary Users

5.1 General Patients and Citizens

People seeking:

- health information;
- preventive-health information;
- guidance on what type of care may be appropriate;
- facility/provider navigation;
- healthcare appointments;
- teleconsultation;
- laboratory services;
- medication-related services where supported.

5.2 Underserved and Low-Connectivity Users

Users who may rely on:

- feature phones;
- IVR;
- USSD;
- SMS;
- low-bandwidth connections;
- voice interaction rather than text.

These users will primarily access PugaAI Health through PugaAccess.

5.3 Existing PugaCare Users

Patients already registered within Puga TriniCare.

5.4 Community Health and Healthcare Users

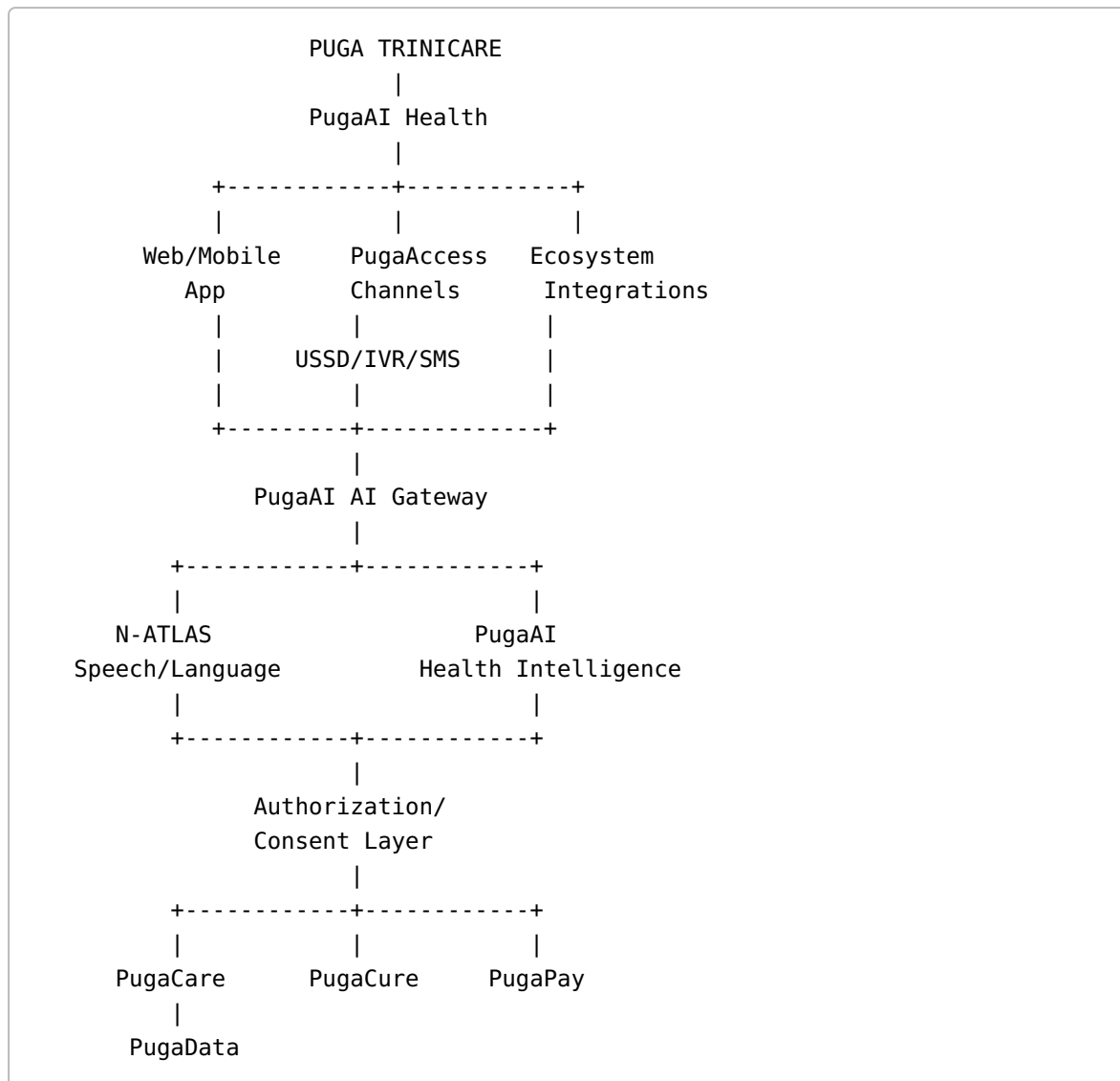
Future versions may support authorized community-health workers and other healthcare personnel.

6. Product Architecture

PugaAI Health shall have its own user-facing access point.

It must not be implemented merely as a feature buried inside PugaCare or PugaAccess.

The architecture shall support:



7. Relationship With Other Puga Products

7.1 PugaAI Health

Primary job:

Understand, inform, navigate and assist.

7.2 PugaAccess

Primary job:

Provide alternative access channels to Puga services for users with limited connectivity, feature phones or low digital literacy.

PugaAccess is not the PugaAI Health product itself.

It is an access/distribution layer through which PugaAI Health can be reached.

7.3 PugaCare

Primary job:

Patient-facing healthcare access and service management.

PugaAI Health can hand off users to PugaCare for:

- provider discovery;
- facility discovery;
- appointment booking;
- teleconsultation;
- laboratory services;
- prescription/refill workflows where supported.

7.4 PugaCure

Primary job:

Healthcare provider and facility workflow.

PugaAI Health may initiate authorized workflows that eventually reach PugaCure.

7.5 PugaPay

Primary job:

Healthcare payments and transactions.

PugaAI Health may initiate a payment workflow through PugaPay where a service requires payment.

7.6 PugaData

Primary job:

Governed analytics and health intelligence.

PugaAI Health interaction data must only enter PugaData through approved privacy, consent, security and data-governance processes.

8. Puga Universal Health ID Integration

PugaAI Health shall use the existing **Puga Universal Health ID** architecture.

A registered user shall have one persistent Puga identity across the ecosystem.

The same identity can be used across:

- PugaAI Health;
- PugaCare;
- PugaAccess;
- participating PugaCure facilities;
- other authorized Puga services.

8.1 Puga Universal Health ID Card

The existing downloadable Puga Universal Health ID Card shall remain supported.

The card contains a QR code associated with the patient's Puga Universal Health ID.

A participating PugaCure facility may:

1. scan the patient's QR code; or
2. search for the patient's Puga Universal Health ID.

The system must then perform appropriate authentication, authorization and consent checks before exposing health information.

8.2 Important Security Principle

The Puga Universal Health ID is an identity identifier.

It must **not itself contain or expose the patient's medical record**.

The flow must be:

```
Puga Health ID
↓
Identity Resolution
↓
Authentication
↓
Authorization
↓
Consent/Access Policy
↓
Permitted Data
```

9. Existing Patient Data-Access Transparency

Puga's current MVP capability:

"Who Has Access to My Health Information"

shall be retained and integrated with PugaAI Health.

The patient should be able to see appropriate information about:

- which facility/provider has access;
- what type of information was accessed;
- when access occurred;
- access status;
- other applicable audit information.

This capability is a core trust and transparency feature.

10. Core Product Capabilities

10.1 Voice Interaction

Users shall be able to:

- start a voice conversation;
- speak naturally;
- ask health questions;
- receive spoken responses;
- receive text responses where appropriate;
- continue a conversation;
- ask follow-up questions;
- end and restart conversations.

The system should minimize the need for users to use medical terminology.

11. Multilingual Interaction

The NAIC MVP shall support N-ATLAS-supported language capabilities that are technically available and appropriate for the implementation.

Initial language support should prioritize:

- Nigerian English;
- at least one validated Nigerian-language experience where feasible.

The architecture must be extensible to:

- Yoruba;
- Hausa;
- Igbo;
- additional Nigerian languages supported by future model capabilities.

Language selection may be:

- explicit;
- automatically detected where reliable;
- configurable in user settings.

12. Health Information

PugaAI Health shall provide structured primary-health information covering selected approved knowledge domains.

Initial domains may include:

- maternal health;
- child health;
- immunization;
- malaria prevention and awareness;
- nutrition;
- hypertension awareness;
- diabetes awareness;
- infectious disease prevention;
- reproductive and family health education;
- general preventive health;
- health-service navigation.

The knowledge base must be curated and version-controlled.

13. Health Navigation

PugaAI Health shall convert health-information conversations into appropriate next steps.

Example:

```
User describes concern
      ↓
PugaAI understands intent
      ↓
Provides appropriate information
      ↓
```

Identifies whether professional care may be appropriate
↓
Offers healthcare navigation
↓
PugaCare
↓
Provider/facility

The system must not represent itself as independently diagnosing the patient.

14. Care Navigation

Where appropriate, PugaAI Health shall allow the user to:

- find a participating facility;
 - find a provider;
 - view available services;
 - initiate an appointment;
 - initiate teleconsultation;
 - navigate to laboratory services;
 - navigate to medication/refill services;
 - access other supported Puga TriniCare services.
-

15. PugaCare Handoff

PugaAI Health shall support a seamless handoff to PugaCare.

Example:

User: "I want to speak to a doctor."

PugaAI Health:

1. identifies the user's request;
2. explains available options;
3. obtains confirmation;
4. transfers the user into the appropriate PugaCare workflow;
5. passes only necessary authorized context.

The user should not have to create a second account.

16. PugaCure Handoff

Where a patient is receiving care at a participating PugaCure facility, PugaAI Health may initiate or support authorized workflows.

Examples:

- appointment-related information;
- patient education;
- follow-up communication;
- service navigation;
- authorized patient information retrieval.

Clinical decisions remain the responsibility of appropriately authorized healthcare professionals.

17. PugaAccess Integration

PugaAI Health shall be accessible through PugaAccess.

Supported future/target channels include:

- IVR;
- USSD;
- SMS;
- low-bandwidth interfaces;
- voice channels.

Example:

```
graph TD; A[Feature phone] --> B[PugaAccess]; B --> C[Voice/IVR]; C --> D[N-ATLAS ASR]; D --> E[PugaAI Health]; E --> F[Response]; F --> G[Voice/SMS];
```

Feature phone
↓
PugaAccess
↓
Voice/IVR
↓
N-ATLAS ASR
↓
PugaAI Health
↓
Response
↓
Voice/SMS

PugaAccess must not create a separate PugaAI identity.

The authenticated user's Puga Universal Health ID should be used where the workflow requires authenticated services.

18. Dedicated PugaAI Health Application

PugaAI Health shall have its own user-facing application.

Initial Access

The product should support:

- responsive web application;
- mobile-friendly/PWA experience.

A native Android/iOS application may follow depending on product validation and development priorities.

Suggested Primary Navigation

- Home
- Talk to PugaAI
- Health Information
- Care Navigation
- My Puga Health ID
- My Conversations
- My Health Journey
- Services
- Notifications
- Privacy & Access
- Settings

19. Home Screen

The primary CTA should be:

Talk to PugaAI Health

Secondary actions:

- Ask a health question;
- Find care;
- View Puga Health ID;
- Find a facility;
- My health information;
- Recent conversations.

The interface should be voice-first rather than text-heavy.

20. Conversation Experience

The conversation interface should contain:

- microphone control;
- text input fallback;
- language indicator;
- conversation transcript;
- voice playback;
- stop/replay control;
- emergency/urgent-care guidance where appropriate;
- escalation to healthcare services where supported.

The interface should clearly distinguish:

- AI-generated information;
 - healthcare-provider information;
 - confirmed patient-record information.
-

21. AI Safety and Clinical Guardrails

PugaAI Health must not:

- claim to be a doctor;
- independently prescribe medication;
- independently make a definitive diagnosis;
- instruct users to ignore urgent symptoms;
- fabricate medical records;
- fabricate provider availability;
- fabricate appointments;
- claim that an action has been completed when it has not.

The system should recognize potential emergency scenarios and direct users toward appropriate urgent professional care or emergency services.

For consequential healthcare actions, the system should use human/provider confirmation where required.

22. AI Architecture

The AI architecture should separate:

A. Speech recognition

N-ATLAS ASR for the NAIC voice implementation.

B. Language/intent understanding

PugaAI processing layer.

C. Health knowledge retrieval

Approved health-information knowledge base.

D. Safety/clinical policy layer

Rules and guardrails that constrain unsafe outputs.

E. Navigation engine

Determines available Puga TriniCare service pathways.

F. Action layer

Initiates authorized actions such as:

- facility search;
- appointment initiation;
- service request;
- payment initiation.

G. Human approval layer

Required for consequential clinical or irreversible actions.

23. AI Response Architecture

The system should follow:

```
graph TD
    A[User Voice] --> B[N-ATLAS ASR]
    B --> C[Intent + Context]
    C --> D[Safety Check]
    D --> E[Health Knowledge Retrieval]
    E --> F[PugaAI Response Generation]
    F --> G[Response Validation]
    G --> H[Voice/Text Response]
```

24. Patient Privacy and Data Protection

PugaAI Health will process sensitive health information.

The product must therefore implement appropriate:

- authentication;
- authorization;
- consent controls;
- role-based access;
- audit logging;
- encryption;
- data minimization;
- secure API communication;
- session management;
- access revocation;
- retention policies;
- incident handling.

PugaAI must not automatically receive unrestricted access to a patient's entire health record.

Only the minimum necessary authorized information should be exposed to an AI workflow.

25. AI and Patient Data Separation

The architecture should distinguish:

1. Identity data
2. Authentication data
3. Health records
4. Consent/permissions
5. AI conversation data
6. Analytics data

Raw identifiable patient information must not be casually used to train AI models.

Any future model-training use must follow approved legal, ethical, privacy and data-governance processes.

26. User Registration

A new user shall be able to register from PugaAI Health.

Registration should support the existing Puga identity architecture.

After successful registration, the user can obtain/access their:

Puga Universal Health ID

and downloadable:

Puga Universal Health ID Card

The user should not need separate registration for PugaCare or PugaAccess.

27. Authentication

The implementation should support secure authentication appropriate to the channel.

Potential mechanisms:

- phone number;
- email where applicable;
- OTP;
- password/passkey where appropriate;
- secure PIN for low-bandwidth channels;
- other approved identity verification mechanisms.

The final mechanism should be aligned with the existing Puga authentication architecture.

28. Facility/Provider Interaction

Participating PugaCure facilities should be able to:

- search Puga Universal Health ID;
- scan the patient's QR code;
- verify identity;
- request/obtain appropriate consent;
- access permitted patient information;
- document the encounter;
- provide authorized follow-up information.

The facility should not automatically receive unrestricted access merely because the patient presents a valid Puga ID.

29. PugaAI Health Entry Points Across the Ecosystem

The product should have its own application while also being accessible from other Puga products.

PugaCare

Button:

Ask PugaAI Health

Purpose:

Patient health information and care navigation.

PugaCure

Button:

PugaAI Assistant

Purpose:

Provider-context AI assistance, subject to role and clinical governance.

PugaAccess

Button/voice option:

Talk to PugaAI Health

Purpose:

Voice/USSD/IVR/SMS access.

PugaAI Health

Primary CTA:

Talk to PugaAI Health

Purpose:

Full consumer AI health experience.

All entry points must connect to the same underlying PugaAI Health service rather than creating separate AI implementations.

30. Contextual Handoff

When a user enters PugaAI Health from another Puga product, authorized context may be passed to the AI service.

Example:

PugaCare facility page:

Facility: XYZ Medical Centre
Services: Consultation, Laboratory

User clicks:

Ask PugaAI Health

PugaAI may receive authorized context that the user is currently viewing that facility.

It should not receive unrelated health records unless explicitly authorized and necessary.

31. Notifications

Future versions may support:

- appointment reminders;
- follow-up reminders;
- medication reminders;
- health education notifications;
- service-status notifications;
- consent/access notifications;
- relevant PugaAI Health alerts.

Notifications must be configurable by the user.

32. Conversation History

Authenticated users should be able to view their PugaAI Health conversation history where retention is enabled.

The system should distinguish:

- AI conversation;
- formal clinical record;
- provider note;
- patient-generated information.

An AI conversation must not automatically become part of the official clinical record.

33. Monetization Requirements

PugaAI Health should initially use a **freemium/access-led model**.

Free

Basic:

- health questions;
- health education;
- basic navigation;
- facility discovery;
- Puga ecosystem navigation.

Transactional Revenue

PUGA may generate revenue when users initiate paid services such as:

- teleconsultation;
- laboratory services;
- medication/refill services;
- other supported healthcare transactions.

PugaPay will handle applicable payment workflows.

Premium

Future premium services may include:

- personalized health support;
- chronic-care support;
- family health management;
- advanced health journeys;
- enhanced reminders;
- premium navigation services.

B2B/B2G

Future enterprise deployment may include:

- government health programmes;
 - PHC networks;
 - hospitals;
 - HMOs;
 - employers;
 - NGOs;
 - telecom partnerships;
 - API licensing.
-

34. NAIC MVP Scope

The NAIC version must remain narrowly focused.

Required NAIC MVP

Must Have

1. Dedicated PugaAI Health interface.
 2. Voice interaction.
 3. N-ATLAS ASR integration.
 4. N-ATLAS language/model integration as required by the selected NAIC challenge.
 5. Primary-health information capability.
 6. Health navigation.
 7. At least one validated Nigerian-language capability where technically feasible.
 8. Puga Universal Health ID authentication architecture.
 9. Basic PugaCare navigation/handoff.
 10. PugaAccess integration pathway.
 11. Interaction logging.
 12. Safety/health guardrails.
 13. Demonstrable real-world user workflow.
 14. Documentation of at least 50 real user interactions for NAIC validation.
-

35. NAIC MVP Exclusions

The first NAIC build should NOT attempt to implement the entire Puga TriniCare ecosystem.

Do not make the following mandatory for the first NAIC build:

- full autonomous clinical decision-making;
- autonomous diagnosis;
- autonomous prescribing;
- complex agentic workflows;

- complete PugaPay integration;
 - complete PugaHSE integration;
 - advanced predictive analytics;
 - nationwide deployment;
 - complete native mobile applications if a responsive/PWA solution is sufficient;
 - large-scale model fine-tuning unless specifically selected as the NAIC challenge route.
-

36. Commercial Roadmap

Phase 1 — NAIC Prototype

Focus:

Voice → N-ATLAS → PugaAI Health → Primary-health information → Navigation

Target:

50+ documented real interactions.

Phase 2 — Public Beta

Add:

- stronger multilingual capability;
- Puga Universal Health ID;
- PugaCare integration;
- PugaAccess IVR;
- user accounts;
- conversation history;
- facility navigation.

Phase 3 — Transactional Health Access

Add:

- appointment booking;
- teleconsultation;
- laboratory workflows;
- medication/refill workflows;
- PugaPay.

Phase 4 — Ecosystem Intelligence

Add:

- personalized health journeys;
- reminders;

- recommendation engine;
- predictive intelligence;
- provider-context assistance;
- authorized agentic workflows.

Phase 5 — Enterprise Platform

Add:

- government deployment;
- enterprise API;
- telecom integrations;
- health-system integrations;
- PugaGov;
- large-scale analytics.

37. Future AI Capabilities

The long-term PugaAI architecture may include:

Predictive Intelligence

Potential predictions:

- appointment no-show risk;
- medicine stock-out risk;
- facility demand;
- revenue trends;
- workforce requirements;
- patient engagement risk;
- operational bottlenecks.

Recommendation Engine

Provide:

- evidence/context-backed recommendations;
- prioritized actions;
- explanations;
- appropriate human approval.

Agentic Workflows

PugaAI may eventually:

1. detect an issue;
2. analyze relevant data;
3. prepare a recommended action;
4. request user/provider confirmation;

5. execute the authorized action;
6. record the outcome.

These capabilities are future roadmap items and should not unnecessarily expand the NAIC MVP.

38. User Stories

Patient

As a patient, I want to speak to PugaAI Health so that I can receive understandable health information without typing.

Patient

As a patient, I want to use my Puga Universal Health ID across Puga services so that I do not need separate identities.

Patient

As a patient, I want to present my Puga Health ID Card at participating facilities so that authorized providers can identify me and access permitted information.

Patient

As a patient, I want to see who has accessed my health information so that I have transparency over my health data.

Patient

As a patient, I want PugaAI Health to connect me to PugaCare when I need professional care.

Feature-phone User

As a low-connectivity user, I want to access PugaAI Health through PugaAccess so that I can use the service without a smartphone application.

Provider

As an authorized PugaCure provider, I want to identify a patient using their Puga Universal Health ID so that I can access permitted information during care.

Facility

As a participating facility, I want to scan a patient's Puga Health ID QR code so that I can efficiently identify the patient.

39. Core Screens

PugaAI Health Web/Mobile

Screen 1 — Landing/Login

- PugaAI Health branding
- Sign in
- Create account
- Continue where appropriate

Screen 2 — Home

- Talk to PugaAI Health
- Ask a question
- Find care
- My Puga Health ID
- Recent activity

Screen 3 — Voice Conversation

- microphone
- transcript
- AI response
- voice playback
- stop/replay
- language selection

Screen 4 — Health Navigation

- recommended next step
- facility/provider options
- service options
- continue to PugaCare

Screen 5 — Puga Health ID

- Puga Universal Health ID
- digital health card
- QR code
- download/share options

Screen 6 — My Access

- Who has access
- facility/provider
- access type
- access timestamp
- permission status

Screen 7 — Conversation History

- previous conversations
- date
- topic
- continue conversation

Screen 8 — Privacy & Settings

- language
 - notification preferences
 - privacy settings
 - consent/access settings
 - account security
-

40. Technical Requirements

The current Puga technical ecosystem uses React/Vite for UI and Express.js for backend services.

PugaAI Health should align with the established Puga engineering architecture unless the technical team documents a justified alternative.

Frontend

Preferred:

- React
- Vite
- responsive design
- Puga design system
- mobile-first interface

Backend

Preferred:

- Express.js
- REST/JSON APIs where appropriate
- authenticated service-to-service communication

AI Services

Dedicated PugaAI Health service layer.

Responsibilities:

- conversation orchestration;
- N-ATLAS integration;

- prompt/context management;
- health knowledge retrieval;
- safety controls;
- service routing.

Data

Use structured persistent storage for:

- users;
- Puga Universal Health IDs;
- consent;
- permissions;
- conversation metadata;
- service requests;
- audit events.

Actual clinical records remain governed by the existing Puga health-record architecture.

41. API Requirements

The architecture should provide APIs such as:

```
POST /api/v1/auth/register
POST /api/v1/auth/login

GET /api/v1/patient/profile
GET /api/v1/patient/health-id
GET /api/v1/patient/health-card

POST /api/v1/ai/conversations
POST /api/v1/ai/voice/transcribe
POST /api/v1/ai/message
POST /api/v1/ai/response

GET /api/v1/ai/conversations

GET /api/v1/care/facilities
GET /api/v1/care/providers
POST /api/v1/care/navigation

GET /api/v1/access/permissions
GET /api/v1/access/audit

POST /api/v1/puga-care/handoff
POST /api/v1/puga-access/session
```

Exact API naming and implementation should be finalized during architecture design.

42. Audit Logging

The system must log security-sensitive events including:

- login;
- identity verification;
- QR scans;
- patient-record access requests;
- consent events;
- authorization decisions;
- health-information access;
- PugaAI actions;
- service handoffs;
- administrative actions.

Logs should be tamper-resistant and access-controlled.

43. Performance Requirements

The system should:

- provide responsive voice interaction;
- provide clear loading states;
- handle temporary network failures gracefully;
- avoid duplicate submissions;
- support retry mechanisms;
- degrade gracefully when AI services are unavailable.

For voice interactions, the UI should clearly indicate:

Listening → Processing → Responding

rather than appearing frozen.

44. Reliability Requirements

The application must:

- prevent loss of completed transactions;
 - safely retry transient API failures;
 - maintain idempotency for consequential actions;
 - prevent duplicate appointment/payment submissions;
 - provide meaningful error messages;
 - maintain service health monitoring.
-

45. Security Requirements

Minimum security requirements:

- HTTPS/TLS;
 - secure authentication;
 - token/session management;
 - role-based authorization;
 - least-privilege access;
 - encryption at rest where applicable;
 - secure secrets management;
 - audit logging;
 - input validation;
 - API rate limiting;
 - protection against prompt injection and malicious input;
 - secure AI tool permissions;
 - separation between AI conversation and clinical record systems.
-

46. AI Security

PugaAI Health must treat external/user-provided content as untrusted.

The AI layer must not be allowed to:

- bypass authorization;
- retrieve arbitrary patient records;
- execute unrestricted backend actions;
- expose system prompts/secrets;
- modify clinical records without authorization;
- perform payments without explicit workflow authorization;
- grant itself additional permissions.

Consequential tools should require explicit permission and appropriate confirmation.

47. Analytics

The initial analytics system should measure:

Usage

- registered users;
- active users;
- voice sessions;
- text sessions;
- repeat users.

Voice

- voice interaction count;
- successful transcription;
- failed transcription;
- language;
- average session duration.

AI

- question categories;
- successful responses;
- escalation rate;
- user feedback.

Navigation

- facility searches;
- provider searches;
- PugaCare handoffs;
- appointment initiation;
- completed service journeys.

Business

- paid service initiation;
- completed transactions;
- revenue generated;
- conversion from AI interaction to healthcare service.

Analytics must follow applicable privacy and data-governance requirements.

48. NAIC Validation Metrics

For the NAIC Voice-First submission, the development team should maintain a validation dataset containing at least:

50 documented real user interactions

Each interaction should capture, where appropriately consented:

- interaction ID;
- date;
- language;
- channel;
- user query;
- ASR result;
- AI response;
- task outcome;
- user feedback;

- errors;
- corrective action.

Personal identifiers should not be included in the public validation evidence unless explicitly authorized and necessary.

49. Success Metrics

NAIC MVP

Success means:

- working PugaAI Health interface;
- genuine N-ATLAS integration;
- functioning voice interaction;
- primary-health use case;
- successful real-user validation;
- documented 50+ interactions;
- demonstrable PugaAccess pathway;
- demonstrable PugaCare navigation pathway.

Commercial Beta

Additional metrics:

- activation rate;
 - weekly/monthly active users;
 - voice completion rate;
 - repeat usage;
 - care-navigation conversion;
 - appointment conversion;
 - teleconsultation conversion;
 - user satisfaction;
 - AI response quality;
 - cost per interaction.
-

50. Definition of Done — NAIC MVP

The NAIC MVP is complete only when:

- ☐ User can register/login.
- ☐ User can access PugaAI Health.
- ☐ User can start a voice conversation.
- ☐ Voice input is processed using the required N-ATLAS integration.
- ☐ PugaAI Health provides a relevant primary-health response.
- ☐ User can ask a follow-up question.

- [] User can receive voice/text response.
 - [] Safety/health guardrails are implemented.
 - [] Puga Universal Health ID architecture is connected.
 - [] PugaCare handoff can be demonstrated.
 - [] PugaAccess pathway can be demonstrated.
 - [] Interaction logging works.
 - [] 50+ real interactions are documented.
 - [] Technical documentation is complete.
 - [] Demo environment works.
 - [] No known critical security vulnerability remains unresolved.
 - [] NAIC demo workflow can be completed end-to-end.
-

51. Out of Scope for NAIC MVP

The following are explicitly deferred unless required for the selected NAIC implementation:

- autonomous diagnosis;
 - autonomous prescription;
 - autonomous clinical decision-making;
 - complete national provider network;
 - complete native iOS/Android apps;
 - full PugaPay integration;
 - advanced PugaData analytics;
 - advanced predictive AI;
 - autonomous agentic workflows;
 - large-scale commercial deployment;
 - unrestricted access to patient medical records by AI.
-

52. Future Product Expansion

PugaAI Health may eventually become the central conversational interface for:

Patients

Health information and care navigation.

Providers

AI-assisted healthcare workflows.

Community Health Workers

Voice-enabled health-information assistance.

Employers

Health and occupational-health navigation through PugaHSE.

Government

Population-health information and citizen-health access through PugaGov.

Health Businesses

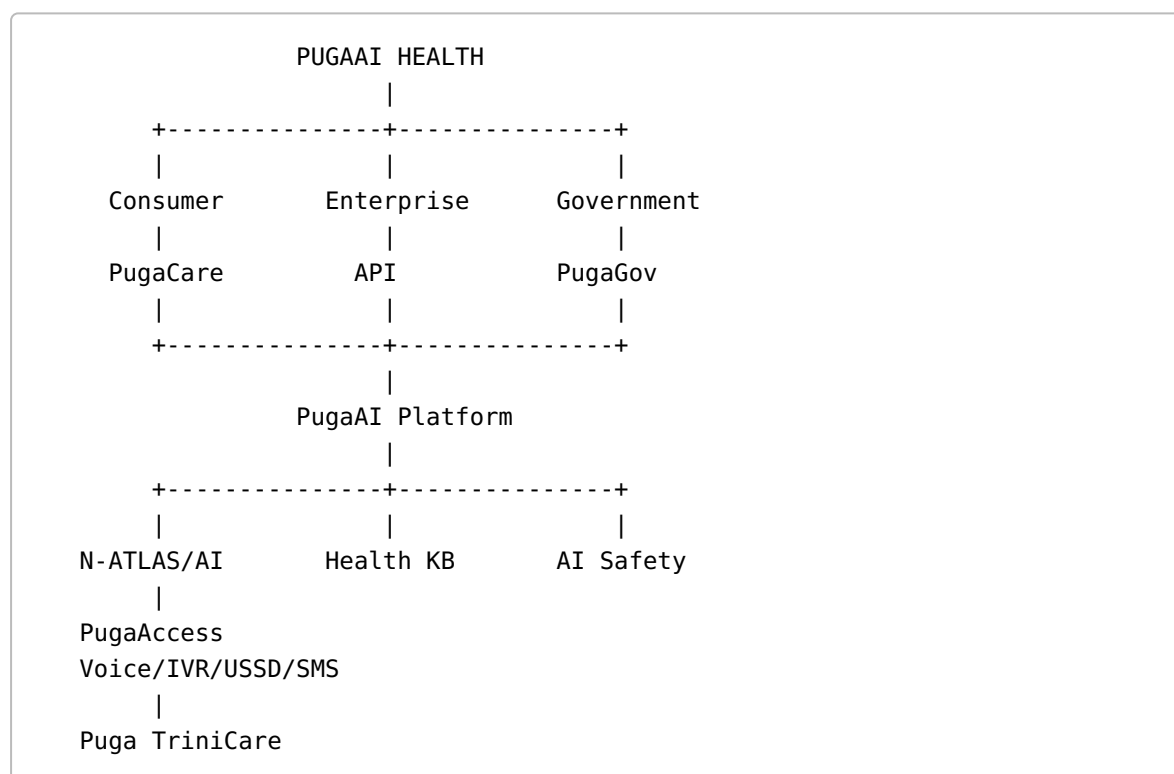
AI/API integration.

Telecoms

Voice-first healthcare access.

53. Commercial Scale Architecture

The long-term platform should support:



The architecture must support horizontal scaling of the AI gateway, API services and access channels without coupling the consumer interface directly to individual AI/model providers.

54. Business Model

PugaAI Health should use multiple revenue channels.

B2C

Free basic health-information and navigation.

Revenue from:

- teleconsultation;
- laboratory services;
- medication/refill transactions;
- premium services;
- other healthcare transactions.

B2B

Revenue from:

- API access;
- enterprise subscriptions;
- healthcare-system deployment;
- implementation;
- support;
- customization.

B2G

Revenue from:

- government deployments;
- PHC programmes;
- population-health programmes;
- citizen-health access platforms;
- analytics/approved intelligence services.

Partnership

Potential partners:

- telecommunications companies;
- HMOs;
- healthcare networks;
- NGOs;
- government agencies;
- pharmacies;
- laboratories.

55. Product Principles

The development team must follow these principles:

1. Voice-first, not voice-only

Voice is the primary interaction paradigm, but text remains available.

2. AI-assisted, not AI-authoritative

PugaAI assists users and providers but does not replace qualified healthcare professionals.

3. One identity

One Puga Universal Health ID across the ecosystem.

4. Consent-first

Patient authorization controls access to protected health information.

5. Minimum necessary data

AI receives only the information required for the task.

6. Human oversight

Consequential healthcare decisions require appropriate human involvement.

7. Ecosystem interoperability

PugaAI Health must integrate rather than duplicate PugaCare, PugaCure and PugaAccess.

8. Channel independence

The same PugaAI Health capability should eventually operate through app, web, IVR, USSD, SMS and other approved channels.

9. Commercial scalability

The architecture must support eventual B2C, B2B and B2G deployment.

10. Build narrowly first

The NAIC MVP should prove the core voice-health proposition before adding complex AI capabilities.

56. Final Product Definition

PugaAI Health — Your AI-powered voice health navigator is the conversational AI health-access layer of Puga TriniCare.

It provides users with a simple way to:

Speak → Understand → Get informed → Navigate → Connect to care → Take action

The dedicated PugaAI Health application provides the full consumer experience.

PugaAccess provides additional low-bandwidth access.

PugaCare provides patient healthcare-service access.

PugaCure enables participating healthcare facilities and providers to deliver care.

PugaPay facilitates healthcare transactions.

PugaData provides governed analytics and intelligence.

The Puga Universal Health ID provides persistent identity across the ecosystem, while consent and authorization controls determine access to protected health information.

The ultimate objective is for PugaAI Health to become the **conversational front door to Puga TriniCare**, while maintaining a strong separation between AI assistance, patient identity, clinical records, healthcare delivery and financial transactions.

57. Developer Priority

Priority 0 — Foundation

- Existing Puga authentication
- Puga Universal Health ID
- authorization/consent integration
- AI service architecture
- N-ATLAS integration
- security/audit foundation

Priority 1 — NAIC Core

- PugaAI Health UI
- voice input
- N-ATLAS ASR
- PugaAI response layer
- primary-health knowledge

- safety guardrails
- voice response
- 50-user validation instrumentation

Priority 2 — Ecosystem Integration

- PugaCare handoff
- PugaAccess integration
- Puga Health ID Card
- patient access transparency
- facility/provider navigation

Priority 3 — Commercial Beta

- appointments
- teleconsultation
- PugaPay
- laboratory navigation
- medication/refill workflows
- notifications

Priority 4 — Scale

- multilingual expansion
- IVR/USSD/SMS scale
- API platform
- enterprise deployment
- government deployment
- predictive intelligence
- recommendations
- agentic workflows

58. Engineering Handoff Statement

The development team should treat this PRD as the product-level definition of PugaAI Health.

Before implementation, engineering must:

1. inspect the existing Puga TriniCare codebase and relevant PugaCare, PugaAccess and PugaCure implementations;
2. identify reusable authentication, patient identity, QR-code, consent, audit and UI components;
3. document existing APIs and integration points;
4. produce the technical architecture;
5. identify any conflicts between this PRD and the current MVP;
6. implement the NAIC MVP before expanding into commercial features;
7. test the complete voice-to-navigation journey;
8. produce evidence for the required real-user validation;
9. conduct security and privacy review before production deployment.

No existing production functionality should be deleted or materially changed without explicit authorization and regression testing.

End of PRD